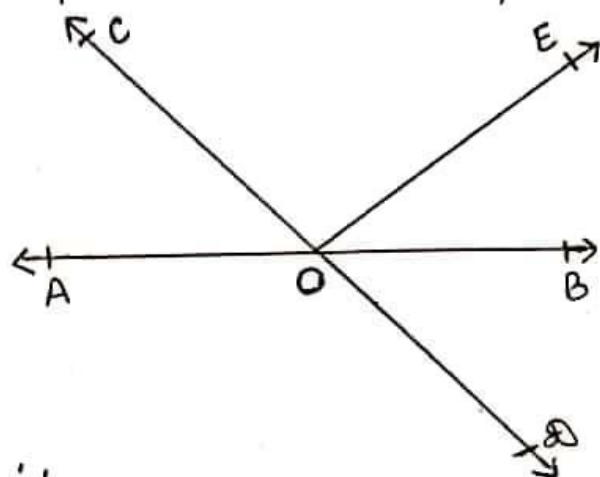


Lines and Angles

Q1 In the figure lines AB and CD intersect at O. If $\angle AOC + \angle BOE = 70^\circ$ and $\angle BOD = 40^\circ$. find $\angle BOE$ and reflex $\angle COE$



Sol: $\rightarrow$ Lines AB and CD intersect at O

$$\angle AOC + \angle BOE = 70^\circ \quad (\text{Given}) \quad \text{--- (i)}$$

$$\angle BOD = 40^\circ \quad (\text{Given})$$

Since $\angle AOC = \angle BOD$ (vertically opposite angles)

$$\therefore \angle AOC = \angle BOD = 40^\circ$$

$$\text{and } 40^\circ + \angle BOE = 70^\circ$$

$$\Rightarrow \angle BOE = 70^\circ - 40^\circ = 30^\circ$$

Also $\angle AOC + \angle BOE + \angle COE = 180^\circ$ ($\because$ AOB is a straight line)

$$\Rightarrow 70^\circ + \angle COE = 180^\circ \quad \text{from (i)}$$

$$\Rightarrow \angle COE = 180^\circ - 70^\circ$$

$$\Rightarrow \angle COE = 110^\circ$$

$$\text{Reflex } \angle COE = 360^\circ - 110^\circ = 250^\circ$$

Hence $\boxed{\angle BOE = 30^\circ}$ and $\boxed{\text{reflex } \angle COE = 250^\circ}$ Ans

Q2 $\Rightarrow$ In the figure, lines XY and MN intersect at O. If $\angle POY = 90^\circ$ and $a:b = 2:3$ find c

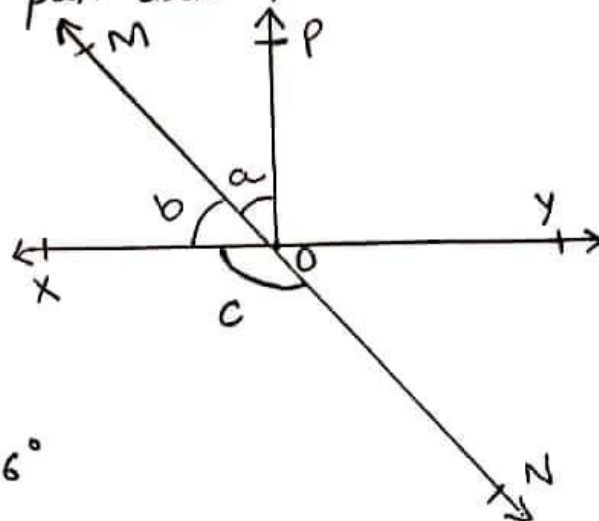
Sol: $\rightarrow$ In the figure line XY and MN intersect at O. ~~If~~ $\angle POY = 90^\circ$

Also given $a:b = 2:3$

Let $a = 2x$, $b = 3x$

Since $\angle XOM + \angle POM + \angle POY = 180^\circ$

(Linear pair axiom)



$$\Rightarrow 3x + 2x + 90^\circ = 180^\circ$$

$$\Rightarrow 5x = 180 - 90^\circ$$

$$\Rightarrow x = \frac{90^\circ}{5} = 18^\circ$$

$$\therefore \angle XOM = b = 3x = 3 \times 18^\circ = 54^\circ$$

$$\text{and } \angle POM = a = 2x = 2 \times 18^\circ = 36^\circ$$

$$\text{Now } \angle XON = c = \angle MOY = \angle POM + \angle POY$$

(vertically opp angle)

$$= 36^\circ + 90^\circ = 126^\circ$$

Hence $\boxed{c = 126^\circ}$ Ans

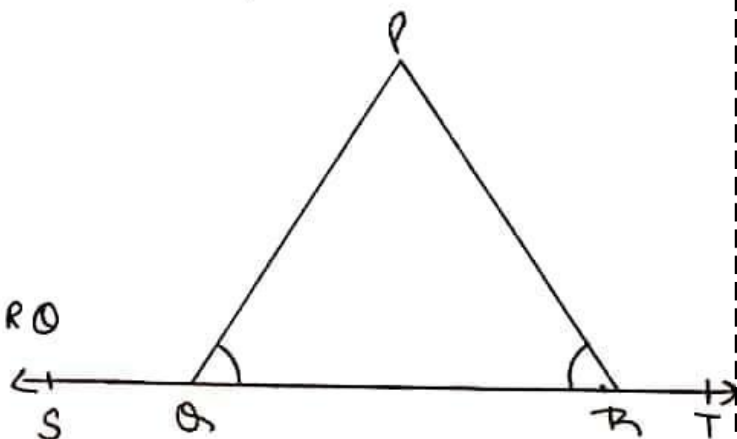
Q3 In the figure $\angle POR = \angle PRO$ then prove that $\angle POS = \angle PRT$

Sol Since $\angle POR = \angle PRO$

$$180^\circ - \angle POR = 180^\circ - \angle PRO$$

$$\Rightarrow \angle POS = \angle PRT$$

Hence proved



Q4 In the figure if $x + y = w + z$ then prove that AOB is a line

Sol: $\rightarrow$ Given $x + y = w + z$

Since $x + y + z + w = 360^\circ$

$\Rightarrow x + y + x + y = 360^\circ$

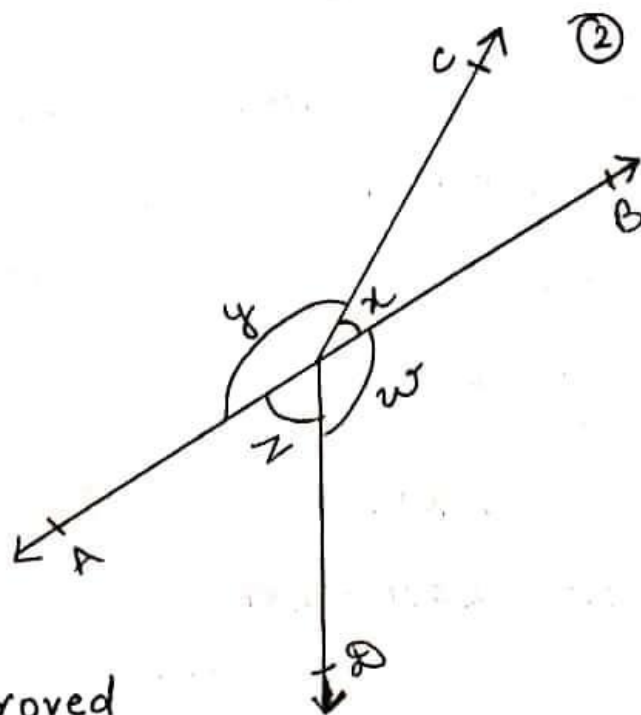
$\Rightarrow 2x + 2y = 360^\circ$

$\Rightarrow 2(x + y) = 360^\circ$

$\Rightarrow x + y = \frac{360^\circ}{2}$

$\Rightarrow x + y = 180^\circ$

$\therefore AOB$ is a Line Proved



Q5: $\rightarrow$ In the figure POQ is a line Ray OR is $\perp$ to line PO . OS is another ray lying between ray OP and OR . Prove that

$$\angle ROS = \frac{1}{2} (\angle QOS - \angle POS)$$

Sol: $\rightarrow \angle ROS = \angle ROP - \angle POS$ — (i)

and $\angle ROS = \angle QOS - \angle QOR$ — (ii)

Adding (i) & (ii) we get

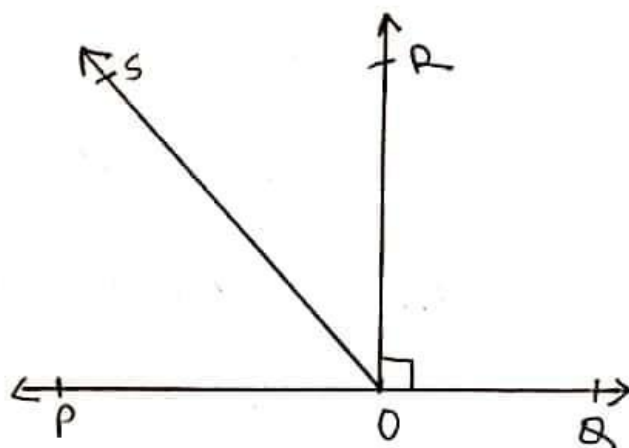
$$\angle ROS + \angle ROS = \angle QOS - \angle QOR + \angle ROP - \angle POS$$

$$\Rightarrow 2\angle ROS = \angle QOS - \cancel{90^\circ} + \cancel{90^\circ} - \angle POS \quad [\because \angle QOR = \angle ROP = 90^\circ]$$

$$\Rightarrow 2\angle ROS = \angle QOS - \angle POS$$

$$\Rightarrow \angle ROS = \frac{1}{2} (\angle QOS - \angle POS)$$

Hence proved //



Q6 $\Rightarrow$ It is given that $\angle XYZ = 64^\circ$ and XY is produced to point P. Draw a figure from the given information.
If ray YO bisects $\angle ZYP$ find $\angle XOY$ and reflex $\angle OYP$

Sol: $\rightarrow$ from figure

$$\angle XYZ = 64^\circ \text{ (Given)}$$

$$\text{Now } \angle ZYP + \angle XYZ = 180^\circ$$

(Linear pair axiom)

$$\Rightarrow \angle ZYP + 64^\circ = 180^\circ$$

$$\Rightarrow \angle ZYP = 180^\circ - 64^\circ = 116^\circ$$

Also given that ray YO bisect $\angle ZYP$

$$\text{But } \angle ZYP = \angle OYP + \angle OYZ = 116$$

$$\Rightarrow \angle OYP = \angle OYZ = \frac{116}{2} = 58^\circ$$

$$\text{Also } \angle XOY = \angle XYZ + \angle OYZ$$

$$\Rightarrow \angle XOY = 64^\circ + 58^\circ = 122^\circ$$

$$\text{and reflex } \angle OYP = 360^\circ - \angle OYP = 360^\circ - 58^\circ = 302^\circ$$

$$(\because \angle OYP = 58^\circ)$$

Hence $\angle XOY = 122^\circ$ and reflex $\angle OYP = 302^\circ$ Ans,,

Exercise 6.2

Q1 : $\rightarrow$ In the figure find the value of x and y then show that $AB \parallel CD$

Sol : $\rightarrow$ In the given figure a transversal intersect two lines AB and CD such that

$$x + 50^\circ = 180^\circ$$

$$\Rightarrow x = 180^\circ - 50^\circ$$

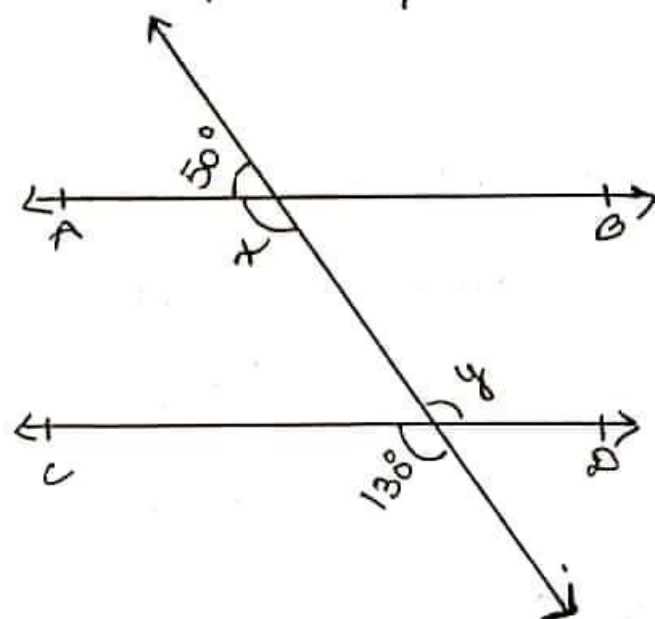
$$\Rightarrow x = 130^\circ \text{ --- (i)}$$

Also $y = 130^\circ$ (vertically opposite angles) --- (ii)

from (i) and (ii)

$$x = y = 130^\circ \text{ (Alternate angles)}$$

$\therefore AB \parallel CD$ (converse of alternate angles axiom)



Q2 : $\rightarrow$ In the figure, if $AB \parallel CD$, $CD \parallel EF$ and $y:z = 3:7$, find x

Sol : $\rightarrow$ In the given figure

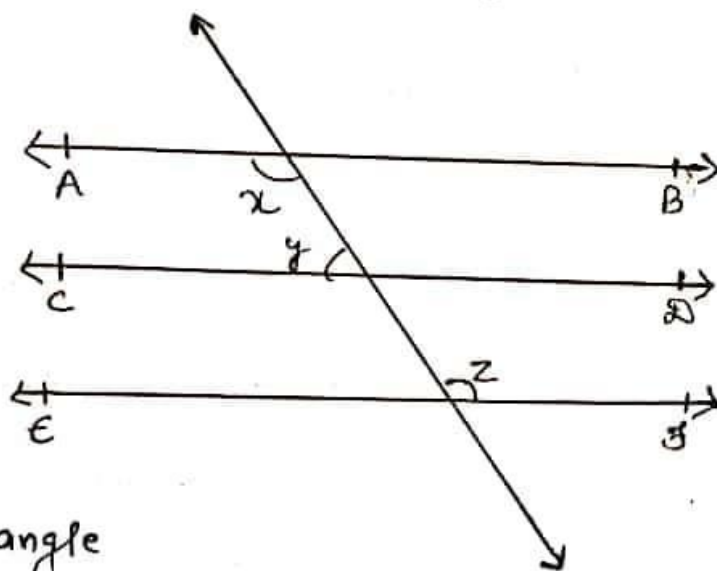
$AB \parallel CD$, $CD \parallel EF$

and $y:z = 3:7$

Let $y = 3a$, $z = 7a$

$\angle DHI = y$ vertically opp angle

$\angle DHI + \angle FHI = 180^\circ$ (Interior angle on same side of transversal)



Q4 : $\rightarrow$ In the figure, if $PO \parallel ST$, $\angle POR = 110^\circ$ and $\angle RST = 130^\circ$
find $\angle ORS$

Sol : $\rightarrow$ Extend PO to Y and draw

$LM \parallel ST$ through R

$\angle TSX = \angle OXS$ (Alternate angle)

$$\Rightarrow \angle OXS = 130^\circ$$

$$\angle OXS + \angle RXO = 180^\circ$$

(Linear pair axiom)

$$\Rightarrow \angle RXO = 180^\circ - 130^\circ = 50^\circ \quad \therefore \angle OXS = 130^\circ$$

$$\text{Also } \angle POR + \angle LRO = 180^\circ$$

(Linear pair axiom)

$$\Rightarrow 110^\circ + \angle LRO = 180^\circ$$

$$\Rightarrow \angle LRO = 180^\circ - 110^\circ = 70^\circ$$

Now

$$\angle ORL + \angle ORS + \angle SRM = 180^\circ$$

$$\Rightarrow 70^\circ + \angle ORS + 50^\circ = 180^\circ$$

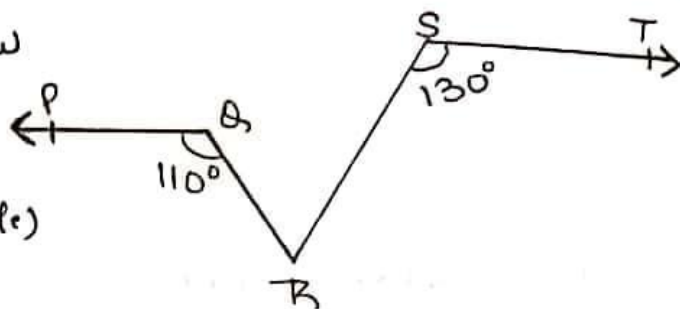
$$\Rightarrow 120^\circ + \angle ORS = 180^\circ$$

$$\Rightarrow \angle ORS = 180^\circ - 120^\circ = 60^\circ$$

$$\Rightarrow \boxed{\angle ORS = 60^\circ}$$

Q5 $\rightarrow$ In figure if $AB \parallel CD$, $\angle APO = 50^\circ$ and $\angle PRD = 127^\circ$
find x and y

Sol : $\rightarrow$ In the given figure $AB \parallel CD$, $\angle APO = 50^\circ$, $\angle PRD = 127^\circ$



$$\Rightarrow y + z = 180^\circ$$

$$\Rightarrow 3a + 7a = 180^\circ$$

$$\Rightarrow 10a = 180^\circ \Rightarrow a = 18^\circ$$

$$\therefore y = 3a = 3 \times 18^\circ = 54^\circ$$

$$\text{and } z = 18 \times 7 = 126^\circ$$

$$\text{Also } x + y = 180^\circ$$

$$x + 54^\circ = 180^\circ \Rightarrow x = 180^\circ - 54^\circ = 126^\circ$$

$$\text{Hence } x = 126^\circ \text{ Ans.}$$

Q3: $\rightarrow$ In the figure, if $AB \parallel CD$, $EF \perp CD$ and $\angle GED = 126^\circ$
Find $\angle AGE$, $\angle GEF$ and $\angle FGE$

Sol: $\rightarrow$ In the given figure $AB \parallel CD$, $EF \perp CD$

$$\text{and } \angle GED = 126^\circ$$

$$\angle AGE = \angle GED \text{ (Alternate angle)}$$

$$\therefore \angle AGE = 126^\circ$$

$$\text{Now } \angle GEF = \angle GED - \angle DEF$$

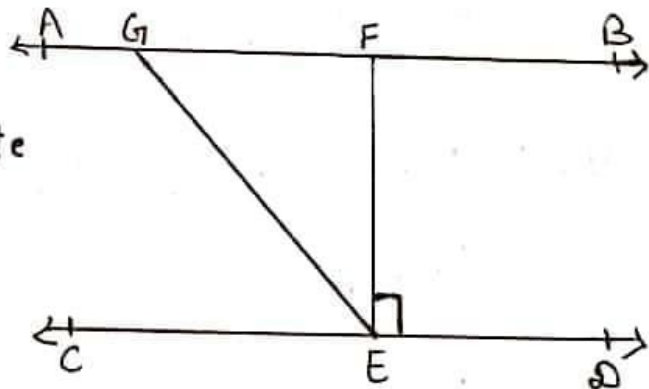
$$= 126^\circ - 90^\circ = 36^\circ (\because \angle DEF = 90^\circ)$$

$$\text{Also } \angle AGE + \angle FGE = 180^\circ \text{ (Linear pair axiom)}$$

$$\Rightarrow 126^\circ + \angle FGE = 180^\circ$$

$$\Rightarrow \angle FGE = 180^\circ - 126^\circ$$

$$\Rightarrow \angle FGE = 54^\circ$$



$$\angle APR = \angle PRD = 127^\circ$$

(Alternate angles)

$$\therefore 50^\circ + y = 127^\circ$$

$$\Rightarrow y = 127^\circ - 50^\circ = 77^\circ$$

$$\Rightarrow \boxed{y = 77^\circ}$$

$$\text{Also } \angle PRD + \angle PRO = 180^\circ$$

(Linear Pair axiom)

$$\Rightarrow 127^\circ + \angle PRO = 180^\circ$$

$$\Rightarrow \angle PRO = 180^\circ - 127^\circ = 53^\circ$$

Now in $\triangle POR$

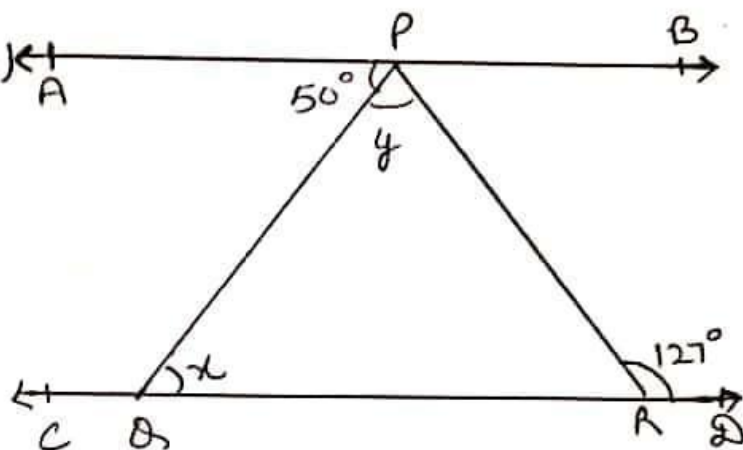
$$x + y + \angle PRO = 180^\circ$$

$$\Rightarrow 77 + x + 53^\circ = 180^\circ$$

$$\Rightarrow x + 130^\circ = 180^\circ$$

$$\Rightarrow x = 180^\circ - 130^\circ = 50^\circ$$

$$\Rightarrow \boxed{x = 50^\circ}$$



Q6 In the figure — — — — Prove that $AB \parallel CD$

Sol At Point B, draw $BE \perp PO$ and at point C draw $CF \perp RS$

$$\angle 1 = \angle 2 \quad \text{---(i)}$$

(Acc to law of reflection)

$$\angle 3 = \angle 4 \quad \text{(same reason) ---(ii)}$$

$$\text{Also } \angle 2 = \angle 3 \quad \text{(Alternate angle) ---(iii)}$$

From (i) (ii) & (iii)

$$L_1 = L_4$$

$$\Rightarrow 2L_1 = 2L_4$$

$$\Rightarrow L_1 + L_1 = L_4 + L_4$$

$$\Rightarrow L_1 + L_2 = L_3 + L_4$$

$$\Rightarrow \angle BCD = \angle ABC$$

Hence $AB \parallel CD$ $\because$ (Alternate angles are equal)

